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Lower knee flexion and hip extension rate of torque development in athletes with previous hamstring strain injury

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ABSTRACT

Persistent deficits in strength and voluntary activation have been observed in athletes with a history of hamstring strain injury. The mechanisms contributing to these deficits are poorly understood and consequently may not be appropriately addressed during rehabilitation. This study aimed to investigate the impact of intended knee flexor contraction mode (concentric, eccentric or isometric) on the rate of torque development and surface electromyography (sEMG) rise in athletes with and without a history of unilateral hamstring strain injury. The impact of the previous injury on hip extensor rate of torque development was also investigated. Previously injured limbs exhibited a slower rate of torque development (mean difference = −31%, p = 0.02, Cohen's d = 0.62) and biceps femoris rate of sEMG rise (mean difference = $-181\% \cdot s^{-1}$, p = 0.003, Cohen's d = 1.10) during intended eccentric knee flexor contractions compared with control limbs. Previously injured (mean difference = −29%, p = 0.01, Cohen's d = 0.85) and contralateral uninjured limbs (mean difference = −31%, p = 0.007, Cohen's d = 0.73) exhibited a slower rate of torque development during isometric hip extensor contractions compared with control limbs. These findings may highlight lower levels of descending input to hamstring motoneurons in previously injured athletes.

ARTICLE HISTORY
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KEYWORDS
Hamstrings; strain injury; neuromuscular inhibition; isokinetic dynamometry; injury prevention

Introduction

Athletes with a history of unilateral hamstring strain injury demonstrate a slower rate of torque development and lower maximal strength during eccentric knee flexor contraction compared with control limbs (Opar et al., 2013b, 2013b). These deficits persist long-term following rehabilitation, and it has been suggested they occur due to an inability to fully voluntarily activate the previously injured limb (Fyfe et al., 2013). The mechanisms contributing to deficits in voluntary activation following hamstring strain injury are poorly understood and as a result, may not be addressed appropriately during rehabilitation. Therefore, understanding whether spinal or supraspinal mechanisms contribute to deficits in voluntary activation following injury may improve rehabilitation outcomes. Determining differences in rapid force production or surface electromyographic (sEMG) activity may indicate lower levels of descending input in previously injured participants, given the relationship between markers of descending drive and explosive force production (Johnson et al., 2015). Additionally, intracortical inhibition induced by transcranial magnetic stimulation limits surface sEMG production during explosive contractions (Irlbacher et al., 2006). Therefore, explosive contractions provide an opportunity to non-invasively investigate whether descending input is altered in athletes with a history of hamstring strain injury.

The rate of torque development in athletes with a prior injury has only been investigated during eccentric contractions (Opar et al., 2013b). In this study (Opar et al., 2013b), there was a discrepancy (100 to 150 ms) between the movement of the dynamometer lever arm and the onset of force production. Therefore, the majority of the contraction used for analysis by Opar and colleagues (2013b) was isometric. This study design provides a unique opportunity to investigate the impact of intended contraction mode on the rate of torque development in previously injured hamstrings. The intention to contract previously injured hamstrings eccentrically may result in lower rates of torque development and sEMG activity compared with concentric or isometric intended contraction modes. If differences between these intended contraction modes exist, it may suggest a need to improve descending input during hamstring contraction throughout rehabilitation.

Much attention has been given to the role of the hamstrings as a knee flexor in the context of strain injury. Previous studies investigating strength differences between participants with and without a history of hamstring strain injury have employed single joint knee flexor contractions (Croisier et al., 2002, 2008; Opar et al., 2013a; Sole et al., 2011). However, while primary movers during knee flexion, the hamstring muscle group also crosses the hip joint posteriorly and has a role in hip extension (Nemeth & Ohlsen, 1985). In healthy young participants, physiological cross-sectional area of the hamstrings far exceeds that of the gluteus maximus (the primary hip extensor) (Handsfield et al., 2014), suggesting a prominent role for the hamstrings in hip extension. Despite this role, the influence of prior hamstring strain injury on hip extensor function has not been investigated.

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The aim of this study was to determine the impact of previous hamstring strain injury on knee flexor rate of torque development and sEMG activity during intended concentric, eccentric and isometric contraction modes. Additionally, we aimed to investigate differences in the rate of torque development of the hip extensors between previously injured and control participants during intended isometric contractions. It was hypothesized previously injured limbs and muscles would exhibit a slower rate of torque and sEMG development during eccentric knee flexor and isometric hip extensor contractions compared with control limbs and contralateral limbs with no injury history.

Methods

Participants

Recreationally active males (n = 20) who participated in several sports (soccer, sprinting, rugby league and rugby union) were recruited for this study. All were free of major lower limb injury at the time of the study. Ten participants (23.4 ± 2.8 years, 182.5 ± 7.4 cm, and 83.0 ± 7.7 kg) had a history of unilateral hamstring strain injury incurred 17.3 ± 6.9 months prior to participation in this study. Injury grade was self-reported by participants based on an assessment from their club or personal physical therapist. The biceps femoris was the involved muscle in all previously injured participants (see Table 1). Ten participants had no history of hamstring strain injury (24.5 ± 3.9 years, 179 ± 6.4 cm, 81.9 ± 9.6 kg), were free of other lower limb injury, and the majority were right-side dominant (n = 9). There was no difference in total weekly training (p = 0.46) or total weekly resistance training time (p = 0.75) between groups. All testing procedures were approved by the Queensland University of Technology Human Research Ethics Committee (approval number: 1600001056) and participants provided written informed consent prior to completing the study.

Table 1. Hamstring strain injury details for participants recruited as part of the injured group.

Participant	Injury site	Grade	Rehabilitation time (weeks)	Time since injury (months)
1	BFlh (LL)	2	6	18
2	BFlh (RL)	2	6	9
3	BFlh (LL)	2	8	4
4	BFlh (RL)	2	5	24
5	BFlh (RL)	2	8	24
6	BFlh (RL)	2	5	24
7	BFlh (RL)	2	8	24
8	BFlh (RL)	2	10	18
9	BFlh (RL)	2	5	18
10	BFlh (RL)	2	6	3

Rehabilitation time was defined by return to pre-injury level of competition. RL = right leg; LL = left leg; BFlh = biceps femoris long head.

Study design and overview

Participants attended two familiarisation sessions involving isokinetic dynamometry. These sessions occurred approximately 1 week prior and were identical to the data collection session. Prior to dynamometry, participants completed a warm-up of 5 minutes of self-paced cycling (50 W) on a Monark bike. Following a warm-up, participants were seated in the dynamometer to perform ballistic contractions. The rate of torque development was first assessed during isometric knee flexor contractions, followed by intended concentric and eccentric knee flexor contractions, and finally, isometric hip extensor contractions. The order of concentric and eccentric contractions was randomised between participants. Following knee flexor contractions, the isometric hip extensor rate of torque development was assessed. Both limbs were tested in a single session with the order randomised between participants.

Surface EMG

Surface EMG activity of the hamstrings was recorded via surface electrodes placed on the muscle belly of the biceps femoris and medial hamstring muscle groups. Following shaving and abrasion, two circular pre-gelled Ag/AgCl electrodes (10 mm diameter, 25 mm inter-electrode distance) were placed halfway between the ischial tuberosity and the medial or lateral epicondyle of the tibia for the medial hamstrings and biceps femoris muscles, respectively (in accordance with SENIAM guidelines). Torque and electromyography data were sampled at 2 kHz through a Powerlab 16/35 recording system, and sEMG was amplified through a quad BioAmp (FE234, ADInstruments).

Isokinetic dynamometry

A Biodex systems 4 dynamometer (Biodex medical systems, Shirley, NY) was used to assess knee flexor and hip extensor rate of torque development. For all contractions, the ankle was secured to the lever arm with the bottom portion of the lever arm reversed so participants were pulling into the less compliant Velcro strapping. During knee flexor contractions, participants were seated upright with the hip flexed by ~80°. Restraints were also applied to the knee of the limb being tested, both hips and across both shoulders to limit unwanted movement (Figure 1(a)). Prior to the testing protocol, participants completed two sets of continuous knee extensor and flexor contractions (range of motion was between 30° and 10° of knee flexion), separated by 2 minutes rest, of six repetitions at $+180° \cdot s^{-1}$. The initial set was a warm-up and the second was used to collect sEMG data for the calculation of the rate of sEMG rise (see below).

One minute following warm-up sets, participants undertook isometric knee flexor, followed by intended concentric or eccentric knee flexor contractions and finally isometric hip extensor rate of torque development contractions (each contraction mode was separated by 2 minutes rest). To initiate contractions, the investigator instructed participants to contract on "go" following a "3, 2, 1, go" instruction. During all contraction modes, participants were shown their torque trace (to minimise pre-tension and provide motivation for

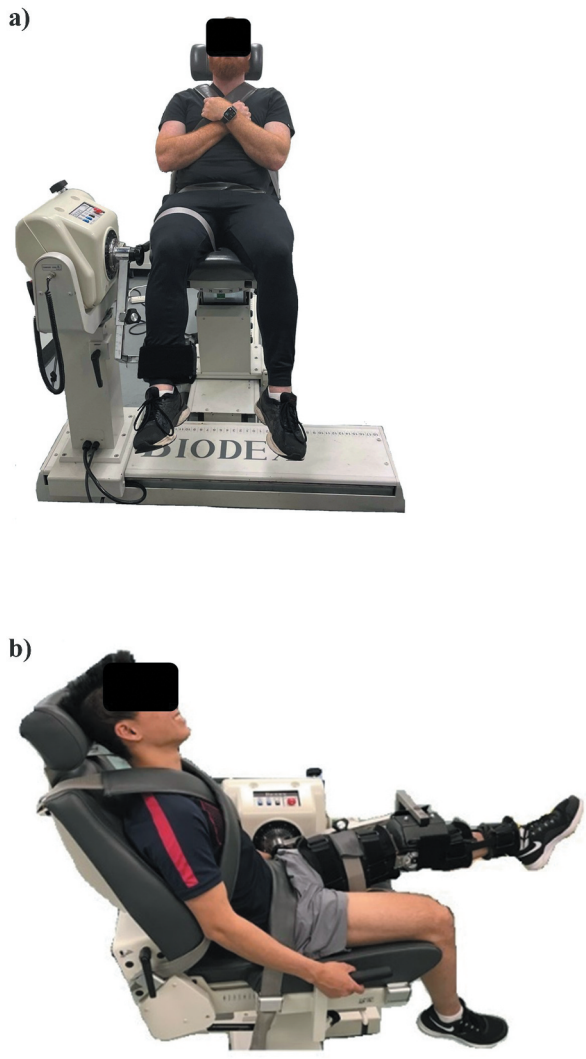

Figure 1. Participants seated on the dynamometer during knee flexion (a) and hip extension (b) contractions. Restraints are applied to the knee, hip and shoulders to limit unwanted movement. A knee brace was placed on participants during isometric hip extensor contractions to limit movement of the knee.

contractions) (Aagaard, 2003) and encouraged to contract "as hard and fast as possible" with emphasis placed on contracting quickly (maximal rate of torque development was the goal during these contractions). Any contraction preceded by countermovement (quadriceps or hip flexor contraction) or an unstable baseline was rejected and repeated.

During isometric knee flexor contractions, the dynamometer lever arm was fixed at 30° of knee flexion and participants completed six contractions, lasting 1 s and separated by 30 s. Intended concentric and eccentric knee flexor rate of torque development contractions were undertaken following isometric knee flexor contractions. The order of these contraction modes was randomised between participants to control for possible effects of fatigue. During intended eccentric contractions, the range of motion of the dynamometer lever was set between 30 and 10° of knee flexion and lever velocity was set to 60° · s^{-1}. The dynamometer was set to reactive mode (for intended eccentric and concentric knee flexor contractions) and did not move until participants began exerting force (the threshold for lever arm movement is 10% of an arbitrarily set maximum torque level, the threshold here was 30 Nm). The average time between the force onset and movement of the dynamometer lever arm was 0.12 ± 0.01 s during eccentric contractions and 0.12 ± 0.02 s during concentric contractions. During concentric contractions, the lever arm range of motion was set between 30° and 50° of knee flexion and movement velocity was set to 60° · s^{-1}. During isometric hip extensor contractions, participants were seated upright with the seat reclined by 15°. The hip joint was aligned with the centre of the axis of rotation of the lever arm and the knee was extended and placed in a rigid brace to limit unwanted movement (Figure 1(b)). The lever arm was placed 15° above the horizontal and attached immediately below the patella. During hip extensor contractions, participants were instructed to push their heel down as fast as possible.

Data analysis

All torque measurements were corrected for limb weight and the raw trace was used for analysis in order to accurately identify contraction onset (Maffiuletti et al., 2016). The rate of torque development was calculated as the average of the torque-time curve (Δtorque/Δtime) from the onset of contraction to 50 and 100 ms for each repetition (see Figure 2). The onset of contraction was determined as four standard deviations above the baseline of the torque-time curve (baseline sampled from 500 ms prior to each contraction). Onset was determined automatically using LabChart software and confirmed via visual inspection. For each contraction mode, the three repetitions with the highest rate of torque development were selected and averaged to calculate rate of torque development for each participant. Of the six contractions collected, all participants produced at least three contractions with similar rates of torque development (Supplement 1). Torque and sEMG onsets identified using four times the baseline standard deviation were valid and torque onsets using this method occurred earlier than those identified using an absolute threshold of 4 Nm (Supplement 2), which has been used previously for identifying onsets (Opar et al., 2013b).Figure 3(a)Figure 4Figure 3(b)

The raw sEMG signal was bandpass filtered (10–500 Hz) using a finite impulse response filterand then full-wave rectified using the root mean square method. The sEMG trace was smoothed over a 10 ms moving window to allow accurate identification of sEMG onset. Surface EMG onset was determined automatically using LabChart software and defined as four standard deviations above the baseline of the sEMG trace. Surface EMG onsets were confirmed with a visual inspection. Rate of sEMG rise was calculated from the same contractions use to calculate rate of torque development (the three contractions of the six performed for each contraction mode exhibiting the fastest rate of torque development). To calculate rate of sEMG rise, the difference in root mean square sEMG amplitude between onset and each time point was divided by each participant's maximal root mean square sEMG activity (100 ms around the peak) during concentric knee flexion at +180° · s^{-1}. This gave a percentage of maximum sEMG produced in

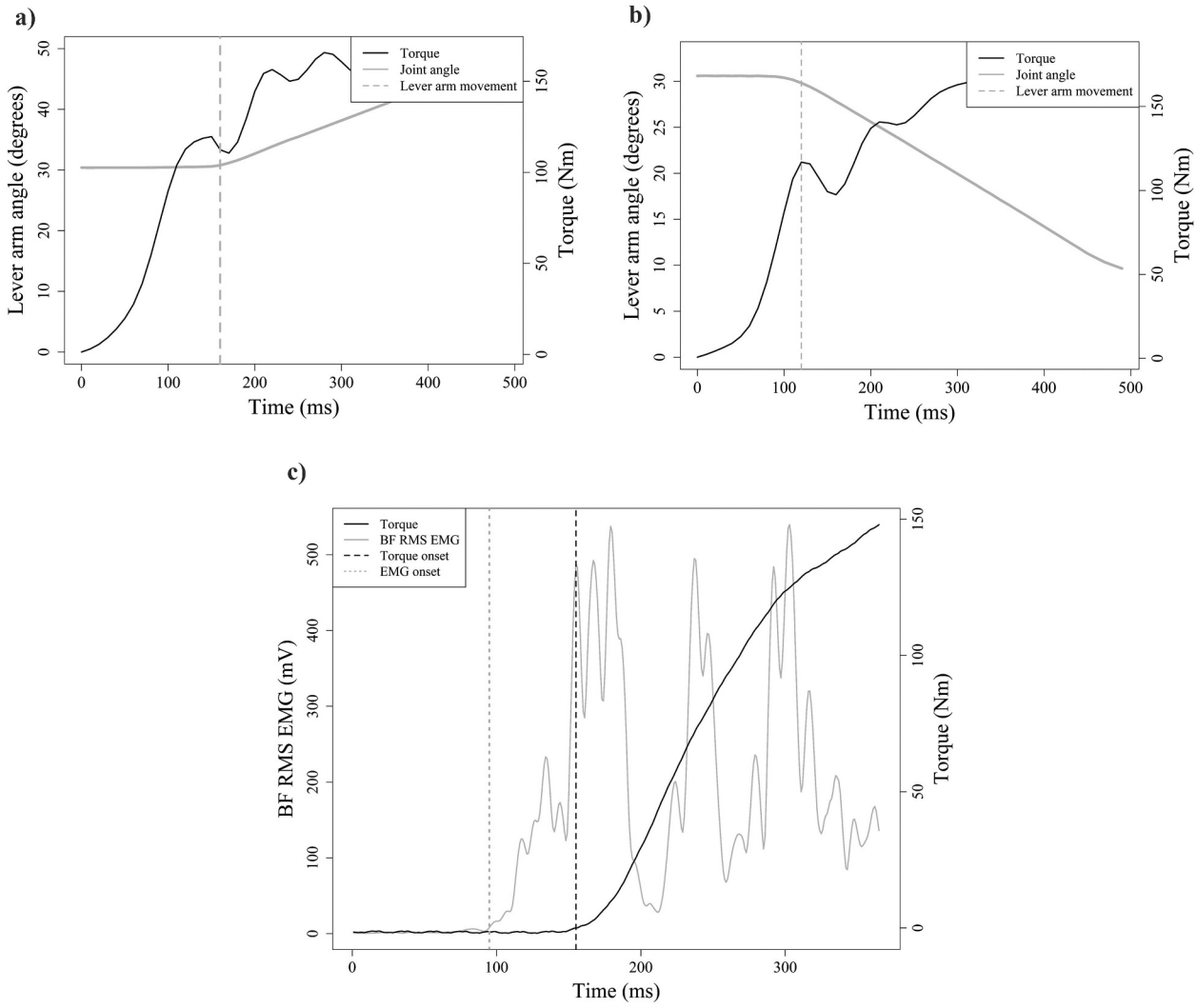

Figure 2. Examples of raw torque-time traces during intended concentric (a) and eccentric (b) knee flexor contractions, where zero on the y-axis represents full knee extension. In panels (a) and (b), zero on the x-axis coincides with the onset of torque production and the vertical dashed line represents the point at which the dynamometer lever arm has moved one degree from its starting position. These panels demonstrate that, despite the intention to contract dynamically, these contractions are isometric. The duration of this initial isometric period is approximately 100–150 ms. Panel (c) demonstrates a torque-time and sEMG-time trace during isometric knee flexor contraction. In panel (c), the zero point does not correspond to torque onset, this is to demonstrate the baseline period prior to contraction and to clearly demonstrate the take-off point of sEMG and torque production. The grey vertical and black vertical lines in panel (c) depict the onset of sEMG activity and torque, respectively.

each time epoch, which was then divided by the corresponding change in time (50 ms or 100 ms), providing the rate of sEMG rise (reported as a percentage of maximum per second).

Statistical analysis

There was no difference in any variable between control dominant and non-dominant limbs. Values for these limbs were averaged (creating a control group average) for comparisons with injured and uninjured limbs in previously injured participants. For each dependent variable, comparisons were made between the control limb average, uninjured limbs of previously injured participants and previously injured limbs. The normal distribution of all dependent variables was confirmed using descriptive methods (skewness and distribution plots) and inferential statistics (Shapiro–Wilk test). Non-normally distributed data (isometric knee flexor rate of torque development, eccentric knee flexor rate of torque development, hip extensor rate of torque development, eccentric knee flexor sEMG rise, concentric knee flexor sEMG rise and biceps femoris hip extensor sEMG rise) were log-transformed for analysis and back-transformed for reporting. Two-way analysis of variance (limb [control, injured, uninjured] × time point [50, 100 ms]) was used to determine differences in the rate of torque development and sEMG rise between limbs for each contraction mode. Where statistically significant effects were observed, Bonferroni's test was used for post-hoc comparisons. Cohen's d values were calculated as a measure of effect size (calculated as $\sqrt{[(s_1 + s_2)/2]}$, where s = standard deviation) with effect sizes interpreted as small < 0.5; moderate 0.5–0.8; large > 0.8. Models were implemented using the statistical software R and packages "emmeans" and "effsize" with α set to 0.05. Correlation coefficients were calculated to determine if the time since injury was related to the rate of torque development or sEMG rise. Time since injury was not related to the rate of torque development (r = 0.06 to 0.41) or sEMG rise (r = 0.01 to

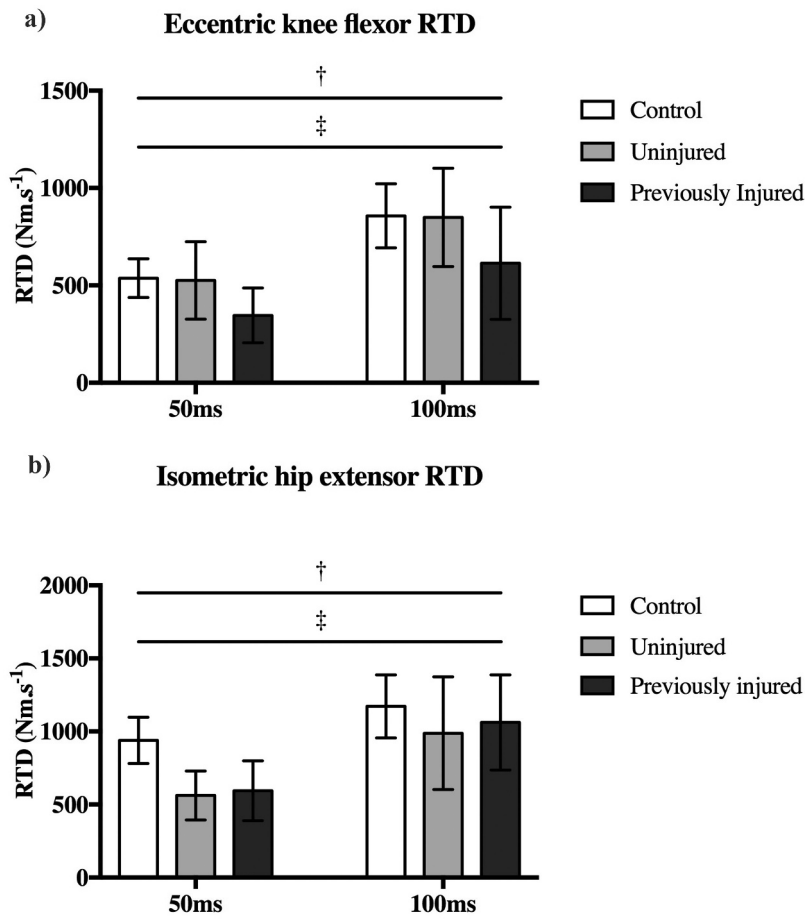

Figure 3. Rate of torque development during eccentric knee flexor (a) and isometric hip extensor (b) contractions. Rate of torque development is displayed on the y-axis, time from contraction onset is displayed on the x-axis. Error bars depict the 95% confidence interval. Dagger (†) and double dagger (‡) symbols denote a main effect of limb and time, respectively. Previously injured limbs exhibited a slower rate of torque development compared with control and contralateral uninjured limbs during anticipated eccentric knee flexor contractions. Previously injured and contralateral uninjured limbs exhibited a slower rate of torque development compared with control limbs during isometric hip extensor contractions. Rate of torque development was faster at 100 ms compared with 50 ms following contraction onset in both contraction modes. RTD = rate of torque development; ms = milliseconds.

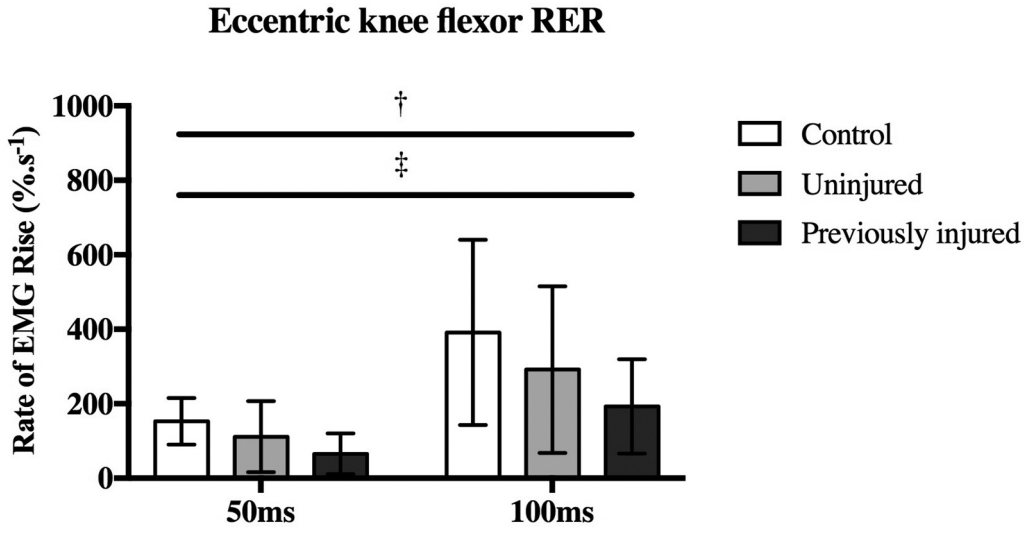

Figure 4. Biceps femoris rate of sEMG rise (RER) during eccentric knee flexor contractions. The y-axis displays the rate of sEMG rise as a percentage of maximal sEMG activity during concentric knee flexion at 180° · s^{-1}, time from contraction onset is displayed on the x-axis. Dagger (†) and double dagger (‡) symbols denote a main effect of limb and time, respectively. Rate of sEMG rise was lower in previously injured compared with control limbs. Rate of sEMG rise was higher at 100 ms compared with 50 ms following contraction onset. Error bars depict the 95% confidence interval, ms = milliseconds.

0.33). Additionally, when included as a covariate in the analysis of variance models, there was no effect of time since injury on any variables (all $p > 0.28$). All data were screened for outliers prior to analysis, no data points were removed. Participant descriptive statistics are reported as means and standard deviations. Mean differences between limbs are reported as the mean and 95% confidence interval.

Results

Impact of limb on rate of torque development

Descriptive statistics for the rate of torque development during each contraction mode and time point are displayed in Table 2. During anticipated eccentric knee flexor contractions, there was a main effect of limb (df = 2, F = 5.99, p = 0.004) and time (df = 1, F = 29.24, p = 0.01) on the rate of torque development (see). Rate of torque development during anticipated eccentric contractions was lower in previously injured compared with control limbs (mean difference = −204 Nm · s^{-1} (−31%), [95% CI = −377 to −32 Nm · s^{-1}], p = 0.02, Cohen's d = 0.62) and contralateral uninjured limbs (mean difference = −187 Nm · s^{-1} (−29%), [95% CI = −363 to −12 Nm · s^{-1}], p = 0.03, Cohen's d = 0.59).

During isometric hip extensor contractions, there was a main effect of limb (df = 2, F = 9.91, p < 0.001) and time (df = 1, F = 34.98, p < 0.001) on rate of torque development (see). Rate of torque development during isometric hip extensor contractions was lower in previously injured compared with control limbs (mean difference = −323 Nm · s^{-1} (−29%), [95% CI = −584 to −62 Nm · s^{-1}), p = 0.01, Cohen's d = 0.85) and in contralateral uninjured limbs compared with control limbs (mean difference = −337 Nm · s^{-1} (−31%), [95% CI = −594 to −80 Nm · s^{-1}], p = 0.007, Cohen's d = 0.73).

During anticipated concentric knee flexor contractions, there was a main effect of time (df = 1, F = 20.24, p < 0.001) on the rate of torque development.

Table 2. Descriptive statistics for rate of torque development during each contraction mode and time point.

Isometric knee flexor contractions (Nm · s^{-1})		
	50 ms	100 ms
Control	600 ± 122	791 ± 157
Uninjured	554 ± 275	753 ± 218
Previously injured	562 ± 163	762 ± 181
Concentric knee flexor contractions (Nm · s^{-1})		
	50 ms	100 ms
Control	567 ± 120	782 ± 169
Uninjured	546 ± 211	744 ± 222
Previously injured	416 ± 142	663 ± 237
Eccentric knee flexor contractions (Nm · s^{-1})		
	50 ms	100 ms
Control	493 ± 86	856 ± 160
Uninjured	507 ± 207	831 ± 272
Previously injured	357 ± 142	634 ± 322
Isometric hip extensor contractions (Nm · s^{-1})		
	50 ms	100 ms
Control	891 ± 170	1414 ± 274
Uninjured	595 ± 305	1010 ± 478
Previously injured	554 ± 178	1060 ± 340

Table 3. Descriptive statistics for rate of sEMG rise during each contraction mode and time point.

Biceps femoris isometric knee flexor contractions (% · s^{-1})		
	50 ms	100 ms
Control	130 ± 59	376 ± 269
Uninjured	117 ± 61	338 ± 275
Previously injured	125 ± 109	316 ± 265
Medial hamstring isometric knee flexor contractions (% · s^{-1})		
	50 ms	100 ms
Control	123 ± 74	370 ± 256
Uninjured	118 ± 48	277 ± 99
Previously injured	99 ± 64	276 ± 155
Biceps femoris concentric knee flexor contractions (% · s^{-1})		
	50 ms	100 ms
Control	372 ± 383	355 ± 225
Uninjured	357 ± 173	328 ± 380
Previously injured	232 ± 136	332 ± 353
Medial hamstring concentric knee flexor contractions (% · s^{-1})		
	50 ms	100 ms
Control	106 ± 75	305 ± 251
Uninjured	106 ± 36	272 ± 127
Previously injured	93 ± 60	262 ± 204
Biceps femoris eccentric knee flexor contractions (% · s^{-1})		
	50 ms	100 ms
Control	219 ± 153	463 ± 269
Uninjured	111 ± 95	291 ± 224
Previously injured	65 ± 54	192 ± 126
Medial hamstring eccentric knee flexor contractions (% · s^{-1})		
	50 ms	100 ms
Control limbs	108 ± 75	318 ± 235
Uninjured	118 ± 41	286 ± 117
Previously injured	82 ± 48	207 ± 118
Biceps femoris isometric hip extensor contractions (% · s^{-1})		
	50 ms	100 ms
Control limbs	103 ± 48	328 ± 204
Uninjured	87 ± 63	219 ± 202
Previously injured	101 ± 98	298 ± 272
Medial hamstring isometric hip extensor contractions (% · s^{-1})		
	50 ms	100 ms
Control limbs	108 ± 44	297 ± 163
Uninjured	87 ± 45	210 ± 126
Previously injured	90 ± 60	184 ± 86

Impact of limb on rate of sEMG rise

Descriptive statistics for the rate of sEMG rise during each contraction mode and time point are displayed in Table 3. There was a main effect of limb (df = 2, F = 11.51, p < 0.001) and time (df = 1, F = 27.33, p < 0.001) on the rate of sEMG rise in the biceps femoris during anticipated eccentric contraction (see). There was a main effect of time on the rate of sEMG rise in the biceps femoris during isometric knee flexor (df = 1, F = 29.56, p < 0.001) and isometric hip extensor (df = 1, F = 21.54, p < 0.001) contractions. There was a main effect of time on the rate of sEMG rise in the medial hamstrings during anticipated eccentric (df = 1, F = 35.96, p < 0.001), isometric (df = 1, F = 30.49, p < 0.001), anticipated concentric (df = 1, F = 34.92, p < 0.001) and isometric hip extensor (df = 1, F = 35.11, p < 0.001) contractions. The rate of sEMG rise in the biceps femoris during anticipated eccentric knee flexor contractions was lower in previously injured limbs compared with control limbs (mean difference = −181% · s^{-1} [95% CI = −308 to −54% · s^{-1}], p = 0.003, Cohen's d = 1.10).

Discussion

This study investigated the impact of intended contraction mode on the rate of torque development and sEMG rise during explosive knee flexor and isometric hip extensor contractions in athletes with and without a history of unilateral hamstring strain injury. The time discrepancy between the onset of contraction and movement of the dynamometer lever arm meant that the first 100 to 150 ms of all contractions were isometric and participants intended to contract isometrically or dynamically during knee flexor contractions. We used this discrepancy to investigate the impact of the injury on torque and sEMG production during different intended contraction modes. The main findings were that athletes with a history of injury exhibited slower rates of torque development and sEMG rise during intended eccentric knee flexor contractions of the previously injured limb compared with their contralateral uninjured limb and control limbs. There were no deficits in torque or sEMG production between participants during other contraction modes of the knee flexors. Both limbs of previously injured participants also exhibited deficits in the rate of torque development during isometric hip extensor contractions compared with control limbs.

It has been suggested that a reduced ability to voluntarily activate the previously injured muscle may contribute to the deficits in the eccentric rate of torque development (Opar et al., 2013b) and strength (Opar et al., 2013a, 2015; Sole et al., 2011) observed following a hamstring injury. The current study expands upon previous findings (Opar et al., 2013b), by demonstrating that deficits in the rate of torque and sEMG development of previously injured limbs are specific to intended eccentric contractions. This suggests that the way eccentric knee flexor actions are planned and executed by those with a history of injury may be different to those without, while rapid intended concentric and isometric knee flexor actions are unaffected. Lower levels of sEMG rise during rapid intended eccentric knee flexor contractions in previously injured compared with control and uninjured limbs suggest an inability to fully voluntarily recruit the previously injured muscle. Although caution is required when inferring voluntary activation from sEMG (Vigotsky et al., 2017), twitch interpolation (a direct measure of voluntary activation) and sEMG assessments seem to be in agreement with one another in the context of hamstring strain injury (Buhmann et al., 2020). Together, slower rates of torque development and sEMG rise suggest an inability to completely activate the previously injured muscle during intended eccentric knee flexor actions. These deficits are likely underpinned by changes at the supraspinal level, given the portion of contractions analysed here were isometric and muscle spindles are unloaded during the initial stages of rapid shortening contractions (Burke et al., 1978). Additionally, participants with afferent neuropathy exhibit similar torque and sEMG traces during explosive contractions compared to those without pathology (Berardelli et al., 1996). Rapid torque production is also strongly related to V-wave amplitude (a marker of descending drive to spinal motoneurons) and the level of recurrent inhibition at rest (the degree of which is controlled centrally) but not the H-reflex amplitude (Johnson et al., 2015). Therefore, rehabilitation strategies aimed at increasing descending drive and limiting recurrent inhibition may be effective in alleviating the long-term deficits in function often observed in participants with a history of hamstring strain injury.

Another novel finding of this study was the slower rate of torque development during rapid hip extensor contractions in both limbs of previously injured compared with control participants. In contrast to knee flexor contractions, there was no deficit in the rate of sEMG rise during hip extensor contractions of either limb in previously injured participants. This may suggest a slower hip extensor rate of torque development is not underpinned by incomplete voluntary activation of the previously injured muscle, or, potential sEMG deficits were not observable from the muscle region sampled here. Alternatively, these deficits may occur due to incomplete activation of other hip extensor muscles that were not assessed with sEMG here, such as the gluteus maximus or adductor magnus. Additionally, given the relationship between muscle volume and force production (Maffiuletti et al., 2016), smaller volumes of other hip extensors in previously injured compared with control limbs may also contribute to deficits in hip extensor rate of torque development. Contribution of hip extensor muscles aside from the hamstrings may also explain the deficiencies in isometric hip extensor contractions that were not observed during isometric knee flexor contractions, as these muscles would have no contribution to knee flexion. In particular, reduced activation or contribution of the adductor magnus to hip extension may explain the deficits observed here, as a prospective study did not identify differences in gluteus maximus sEMG activity during treadmill running in players who did and did not go on to sustain a hamstring injury (Franettovich Smith et al., 2017). Additionally, the moment arm of the adductor magnus increases with the degree of hip flexion (Nemeth & Ohlsen, 1985). This suggests the contribution of this muscle to hip extension increases as the hip becomes more flexed. As hip extensor tests were employed with the hip in ~75° of flexion, where the adductor magnus moment arm peaks (Nemeth & Ohlsen, 1985), there is potential this muscle may contribute to hip extensor deficits observed here.

There are some limitations associated with the methodology of the current study. Surface EMG cannot be used to make direct inferences regarding voluntary activation of muscle (Vigotsky et al., 2017), and therefore it cannot be definitively stated that voluntary activation is incomplete during rapid intended eccentric knee flexion in previously injured participants. Future studies should employ electrical stimulation during rapid knee flexor contractions to determine if voluntary activation is incomplete. Additionally, dynamic contractions may exhibit lower reliability than isometric contractions when the rate of torque development is being analysed (Maffiuletti et al., 2016). To overcome this limitation, participants were required to complete two familiarisation sessions prior to testing. The within-participant coefficient of variation for the rate of torque development for all time points and contraction modes was between 8% and 15%, which is comparable to other values reported in the literature (Maffiuletti et al., 2016). Finally, the retrospective design of this study means we are unable to determine if the deficits observed here are the cause or result of injury.

In conclusion, this study reports a slower rate of torque development and sEMG rise of the biceps femoris during intended eccentric knee flexor contractions in limbs with a history of hamstring strain injury, compared with control and uninjured limbs. This may be reflective of deficits in descending input to hamstring motoneurons. Additionally, both limbs of previously injured participants exhibited a slower rate of torque development during isometric hip extensor contraction. These deficits persist long-term following injury and may require novel rehabilitation strategies.

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